

CloudShieldX

Phase 2 — Consolidated Systematic Literature Review

Research Landscape, Prior Art, Thematic Synthesis & Gap Definition

Version 1.0 | August 2026

Working research direction: DAIL — Dependency-Aware Invariant Ledger

Document Control & Evidence Boundary

Item	Value
Purpose	Master Phase 2 reference consolidating the SLR protocol, collection record, screening, quality assessment, extraction, thematic synthesis and current research-gap conclusion.
Primary research subject	LLM-driven Infrastructure-as-Code security remediation, verification, regression prevention and stateful invariant preservation.
Product context	CloudShield AI remains the broader product vision. DAIL is the current candidate research core inside that product.
Evidence status	Mixed: primary-paper evidence, user-supplied research reports, AI-assisted extraction and selected verified web sources. Evidence boundaries are preserved.
Novelty status	Not proven. DAIL is a strong research-gap candidate, not a confirmed novel or patentable invention.
Phase 2 decision	Proceed to Phase 3, with continued prior-art monitoring and no “first ever” claims.

Executive Summary

The Phase 2 review was designed to test—not confirm—the research direction from Phase 1. The review examined LLM-assisted IaC generation and remediation, Terraform security, verification and validation, functional correctness, regression behavior, formal methods, invariants, CEGAR/CEGIS, and dependency-aware infrastructure analysis.

The literature strongly establishes that the generic pipeline of an LLM generating or repairing Terraform and then receiving verifier feedback is no longer sufficient as a research contribution. Recent systems already combine tool grounding, provider/dependency context, scanner feedback, Terraform validation or planning, policy gates, layered oracles, and sandbox testing. The uploaded thematic synthesis explicitly describes the mechanical pipeline as increasingly saturated. [Source: Phase 2 thematic synthesis, pp. 1–3]

The most important remaining issue is whether an iterative repair process can preserve properties already verified in earlier iterations. The review repeatedly identifies deceptive fixes, resource restructuring, and regression of previously passing controls as central failure modes. [Source: Phase 2 thematic synthesis, pp. 8–9]

The strongest current gap candidate is a Terraform-specific mechanism that maintains persistent verified state across repair iterations, tracks resource/dependency identity, and selectively re-verifies affected properties. This candidate is referred to as DAIL (Dependency-Aware Invariant Ledger).

However, the review does not establish absolute novelty. Adjacent formal-methods work already demonstrates important components such as contract refinement, invariants and counterexample-guided verification. Therefore, DAIL must be framed as a Terraform/IaC-specific regression-control mechanism rather than as the invention of stateful verification itself. [Source: Stateful-invariant full-text extraction, pp. 10–11]

Phase 2 therefore passes as a research-validation stage, but the contribution remains provisional until Phase 3 formalizes the mechanism and additional targeted prior-art/patent checking is performed against the exact algorithm.

1. Review Objective and Scope

The SLR objective was to determine what existing research already provides for LLM-driven Terraform/IaC security remediation and to identify whether a defensible unresolved problem remains around preserving previously verified infrastructure properties across iterative AI repair.

1.1 Review themes

- LLM / GenAI / agentic IaC generation
- LLM-based Terraform security remediation
- Layered verification and oracle design
- Functional correctness and regression evaluation
- Formal verification, SMT/Z3 and model checking
- Security and functional invariants
- CEGAR / CEGIS and counterexample-guided repair
- Dependency graphs and resource identity
- Stateful verification and property preservation

1.2 Review principles

- Do not assume the research gap is valid; search for evidence that would invalidate it.
- Separate primary evidence from AI-generated or secondary summaries.
- Treat “not reported” differently from “no.”
- Do not equate scanner-warning disappearance with correctness.
- Do not claim novelty or patentability without a dedicated prior-art assessment.

2. Search Strategy and Collection Record

The search strategy used a three-tier design: (A) exact intersection, (B) adjacent research, and (C) foundational verification. AI tools were used to expand discovery and assist screening, while the original source of each important claim remained a primary-paper or authoritative source wherever possible.

2.1 Three-tier model

Tier	Purpose	Example queries
A — Exact intersection	High-precision discovery of LLM + Terraform/IaC + verification/remediation.	LLM + Terraform + formal verification; LLM + Terraform + remediation.
B — Adjacent	Find related verification, correctness and repair research even when not all concepts co-occur.	IaC + formal verification; IaC + functional correctness/regression.
C — Foundations	Establish cloud verification, invariants, reachability and repair foundations.	AWS formal verification; CEGAR/CEGIS; security invariants.

2.2 Practical collection record

Source	Recorded evidence	Status / limitation
IEEE Xplore	S1 LLM + Terraform + security = 8; S2 LLM + Terraform + remediation = 2; S3 Terraform + formal verification = 0; S4 Terraform + functional correctness = 0.	Documented four-search pass completed.
ACM Digital Library	S1 LLM + Terraform + security = 93.	Partial execution; later searches not completed.
Scopus	Search interface could not be used successfully.	Not searched; explicitly documented as unavailable.
Web of Science	Clarivate Master Journal List was reached, not the literature-search interface.	No validated Web of Science paper count recorded.
ScienceDirect / SpringerLink	Query strategy prepared, but a complete execution log was not captured.	Not treated as fully executed indexed-database evidence.
Google Scholar / Semantic Scholar	Used for discovery/identification and citation exploration.	Discovery sources, not substitutes for indexed-database evidence.
Elicit	13-paper candidate set and assisted screening.	AI-assisted discovery/screening.
Gemini Deep Research	Citation-network expansion, thematic synthesis and gap challenge.	Secondary evidence; important claims independently checked when possible.

Source	Recorded evidence	Status / limitation
ResearchRabbit / Connected Papers	Planned as network-expansion aids; manual interaction was minimized.	Optional/secondary.
Scite / NotebookLM	Used/planned for citation checking and synthesis.	Secondary analysis tools.

Evidence boundary: the review should be described as a structured, AI-assisted SLR with partial indexed-database coverage plus targeted primary-source verification. It should not be described as a fully exhaustive database search across every planned source.

3. Candidate Screening and Full-Text Verification

The screening process used INCLUDE / EXCLUDE / MAYBE decisions, followed by full-text verification where a paper was a high-priority prior-art threat. The screening stage intentionally kept ambiguous papers in MAYBE status rather than forcing premature exclusions.

3.1 Priority papers and status

Study	Current classification	Evidence status	DAIL relevance
TerraRepair	Include — Direct	Primary/strong source material verified	Direct prior art for tool-grounded Terraform repair and dependency context.
TerraProbe	Include — Direct	Primary/strong source material verified	Direct prior art for layered oracle evaluation and deceptive fixes.
SafeFix-Agent	Include — Direct / Low confidence	Detailed primary evidence incomplete	Relevant gating, diff-only remediation and sandbox/testing concepts.
Verifier-First Evaluation	Include — Adjacent	Primary/strong source material reviewed	Shows verifier-first iterative IaC generation with validate/plan/OPA feedback.
LLMSecConfig	Include — Adjacent	Primary/strong source material reviewed	Important adjacent evidence for security + operational preservation.
IAMPERE	Foundational/Adjacent — pending stronger source verification	Current extraction had insufficient source material	Potential formal cloud repair comparison; do not rely on unsupported details.
Zelkova	Foundational	Authoritative/primary background	Formal AWS policy reasoning foundation.
Tiros	Foundational	Authoritative/primary background	Formal AWS network-reachability foundation.
UPIR	Adjacent — pending primary verification	Earlier extraction lacked sufficient direct source evidence	Stateful/proof-caching inspiration; not a Terraform repair system.
PolyVer	Adjacent	Primary publication/source identified	LLM contract refinement and compositional verification in C/Rust, not Terraform.
ConVer	Adjacent — major threat	Primary source identified	LLM + CEGAR/CEGIS + persistent contract refinement in C.

3.2 Evidence-boundary rule

The review explicitly refused to infer persistent invariant ledgers, resource-identity enforcement, dependency invalidation, or proof caching when the available abstract or report did not state them. The evidence-boundary report emphasizes this conservative approach. [Source: Evidence-boundary report, opening note]

4. Quality Assessment

Studies were scored across six criteria: objective clarity, methodological rigor, empirical validation, regression/correctness measurement, reproducibility, and threats-to-validity/limitations. Each criterion used 1 = yes, 0.5 = partial, 0 = no. The assessment was provisional where only summaries or excerpts were available.

4.1 Provisional quality matrix

Study	Score	Quality	Important caveat
TerraRepair	3.0 / 6	Low-confidence	Scored from supplied evidence; final paper-quality judgment requires full primary-text audit.
TerraProbe	4.5 / 6	Medium	Strong evaluation evidence; limitations not fully established in supplied extraction.
SafeFix-Agent	1.5 / 6	Low-confidence	Detailed methodology/results were not sufficiently available.
Verifier-First	4.0 / 6	Medium	Strong experimental evidence; explicit limitations not fully captured.
LLMSecConfig	5.0 / 6	High	Strong benchmark and comparative evidence in supplied extraction.
IAMPERE	5.0 / 6 provisional	High-confidence only if primary source verified	Current direct extraction later reported search failure; do not use unsupported details.
Zelkova	4.0 / 6	Medium	Foundational production technology rather than a conventional empirical study.
Tiros	3.0 / 6	Low-confidence in current matrix	Important foundation, but empirical details were not all present in the supplied extraction.
UPIR	5.0 / 6 provisional	High-confidence only if primary source verified	Current extraction later marked source availability insufficient; keep as adjacent hypothesis.
PolyVer	5.5 / 6	High	Strong formal-method evidence; domain differs from Terraform.
ConVer	5.0 / 6	High	Strong formal/CEGAR evidence; domain differs from Terraform.

Interpretation warning: these are evidence-weighting scores for the material available during Phase 2, not immutable judgments on the scientific quality of the underlying studies.

5. Literature Extraction Matrix — Key Dimensions

The extraction matrix was designed around 21 fields, with five DAIL-critical dimensions receiving special attention: persistent state, invariant ledger, resource identity, dependency-aware invalidation, and property/proof caching.

Study	LLM + Terraform/laC	Security remediation	Functional verification	Stateful/invariant mechanism	DAIL difference
TerraRepair	Yes	Yes	Partial / scanner-oriented	No persistent ledger; dependency context present	Adds dependency context but not persistent cross-iteration preservation.
TerraProbe	Yes	Evaluation, not repair generation	Plan/behavior/semantic layers	No iterative state; provenance is audit-oriented	Strengthens the oracle but not the repair-state mechanism.
SafeFix-Agent	Likely / not fully established	Yes	Policy + test + sandbox gates	No persistent ledger demonstrated	Transactional gates rather than invariant memory.
Verifier-First	Yes	No; generation study	Validate + plan + OPA	No persistent cross-task state	Verifier-first feedback, but no resource-aware invariant ledger.
LLMSecConfig	LLM + Kubernetes rather than Terraform	Yes	Operational preservation	Iterative re-scanning; no persistent ledger	Adjacent domain; repeated checking rather than Terraform identity-aware state.
PolyVer	LLM + contracts	No laC remediation	Formal	Iterative contract	Strong verification

Study	LLM + Terraform/IaC	Security remediation	Functional verification	Stateful/invariant mechanism	DAIL difference
			contracts/model checking	refinement, not Terraform state	mechanism, different domain.
ConVer	LLM + contracts	No IaC remediation	Formal contracts/loop invariants	Persistent contracts within CEGAR/CEGIS	Strongest methodological adjacent threat; no cloud resource semantics.
Zelkova	No LLM	No repair	Security policy verification	Static formal analysis	Formal AWS security oracle, not iterative AI repair.
Tiros	No LLM	No repair	Network reachability	Static formal analysis	Formal AWS network oracle, not iterative AI repair.

6. Thematic Literature Analysis

6.1 Theme A — LLM + IaC generation and remediation

The literature has moved from unconstrained generation toward tool-grounded remediation. TerraRepair exemplifies this shift by using provider/dependency context and scanner feedback. The broader synthesis describes this as a response to hallucinated resources, schema mistakes and missing deployment context. [Source: Phase 2 thematic synthesis, p. 1]

6.2 Theme B — Layered verification and deceptive fixes

TerraProbe is important because it demonstrates that targeted scanner-warning removal can substantially overstate successful remediation. Its multi-layer approach separates targeted findings, full-scanner cleanliness, planning and semantic/intent validation. [Source: Phase 2 thematic synthesis, pp. 1–3]

6.3 Theme C — Regression and iterative volatility

A major shift in the literature is recognition that cumulative-best metrics can hide destructive per-iteration behavior. The thematic synthesis reports that iterative repair can break previously passing controls and identifies resource restructuring as a major mechanism. [Source: Phase 2 thematic synthesis, pp. 1–2]

6.4 Theme D — Security vs functional verification

Terraform validation and planning answer different questions from security verification. A configuration may be syntactically valid and plan successfully while remaining insecure, and a security-oriented change may alter desired functionality. The literature therefore treats the oracle problem as multidimensional. [Source: Phase 2 thematic synthesis, pp. 3–4]

6.5 Theme E — Formal and compositional verification

Adjacent formal-methods research demonstrates that invariants/contracts and counterexample-guided refinement are viable techniques. ConVer is particularly relevant because it retains synthesized contracts across CEGAR/CEGIS iterations, although it targets C rather than Terraform. [Source: Stateful-invariant full-text extraction, pp. 10–11]

6.6 Theme F — Dependency and resource identity

Terraform-specific systems already use dependency information as context. The current evidence does not yet show a complete mechanism that turns that context into persistent, regression-aware protection of previously verified properties. TerraRepair is especially relevant because its dependency graph is already close to one building block of DAIL. [Source: Stateful-invariant extraction, pp. 1–2]

7. Contradictions and Negative Evidence

Issue	Competing observations	Research implication
Iterative feedback	Cumulative-best metrics imply improvement; per-iteration analyses reveal regressions.	A repair system must be evaluated on trajectory, not only best/final score.
Full rescanning vs stateful ledger	Full rescanning may be computationally cheap; stateful memory may offer semantic protection rather than speed.	DAIL should not rely primarily on proof-cache latency as its motivation.
Security vs functionality	Security checks can pass while operational behavior changes; plan success does not prove security intent.	Joint property preservation must be explicit.
More constraints vs model knowledge	Remembering prior invariants does not supply missing provider/schema knowledge.	DAIL complements, rather than replaces, tool grounding/RAG/provider context.
Sandboxing vs invariant history	A final sandbox can validate final behavior but may not explain or prevent earlier harmful mutations.	A stateful ledger can be evaluated as a regression-control mechanism, not merely a faster validator.

8. State of the Art Synthesis

Area	Current assessment	Evidence status
Static detection / scanner-grounded repair	Mature / saturated as a novelty claim	Strong.
Tool-grounded Terraform agents	Mature and rapidly evolving	Strong direct prior art.
Layered verification / deceptive-fix detection	Emerging to established	Strong TerraProbe evidence.
Security + functional verification	Partially established / fragmented	Direct and adjacent evidence.
Formal cloud security verification	Mature in specific AWS domains	Strong foundational evidence.
Contract / CEGAR / CEGIS with LLMs	Established in adjacent software-verification domains	Strong adjacent evidence.
Persistent invariant ledger for iterative Terraform security remediation	Underexplored in the reviewed evidence	Promising, not proven novel.
Resource-identity-aware regression control for LLM Terraform repair	Underexplored in the reviewed evidence	Promising, needs formalization.
Dependency-aware selective re-verification for iterative Terraform repair	Underexplored in the reviewed evidence	Promising, needs experiments.

9. Research Gap Synthesis

9.1 What is already solved

- LLM-assisted Terraform/IaC repair
- Scanner-grounded remediation
- Provider/schema and dependency context injection
- Terraform validate/plan feedback
- Policy-as-code gates
- Layered oracles
- Sandbox/smoke-test validation
- Formal AWS policy/network verification
- LLM-driven contract refinement in adjacent software domains

9.2 What remains fragmented

The reviewed evidence shows multiple systems covering different pieces of the problem, but no validated source in the current evidence set demonstrates the complete combination of: persistent verified property state, Terraform resource identity protection, dependency-aware invalidation/selective re-verification, and multi-iteration LLM security remediation.

This statement is deliberately narrower than “no one has done this.” It means that within the evidence actually verified during Phase 2, the exact combination was not demonstrated.

9.3 Current gap statement

Provisional research gap: Existing work substantially covers LLM-based Terraform remediation and multi-layer verification, while adjacent formal methods cover invariant/contract refinement. What remains insufficiently demonstrated is a Terraform-specific, regression-aware mechanism that persistently preserves previously verified properties across successive LLM remediation steps while tracking resource/dependency identity and selectively re-verifying affected properties.

10. DAIL — Candidate Research Contribution

DAIL stands for Dependency-Aware Invariant Ledger. It is a candidate architectural contribution, not yet a locked final implementation.

10.1 Conceptual workflow

1. Capture a baseline Terraform configuration and resource/dependency identity.
2. Detect or receive a security finding.
3. Generate a constrained candidate patch with the LLM.
4. Verify the candidate against the new requirement and the stored invariant set.
5. Determine whether resource identity or protected dependencies changed unexpectedly.
6. Re-verify only the affected properties plus required global invariants.
7. Reject or accept the patch; update the ledger only after verification succeeds.
8. Continue until the repair converges or reaches a bounded failure state.

10.2 Candidate mechanisms

Mechanism	Purpose	Why it matters
Persistent invariant state	Record previously verified security/functional properties.	Prevents “forgetting” what was already correct.
Resource identity map	Map Terraform resources/modules to stable identities.	Reduces false regressions and prevents destructive restructuring.
Dependency graph	Identify which properties can be affected by a change.	Enables targeted re-verification.
Selective re-verification	Recheck affected properties rather than treating every edit as unrelated.	Potentially improves regression control and explainability.
Regression-aware acceptance gate	Reject a patch if protected properties become invalid.	Turns historical correctness into an explicit acceptance constraint.

10.3 What DAIL is NOT

- Not a claim that LLM + Terraform + verification is novel.
- Not a new general formal-verification paradigm.
- Not merely a scanner wrapper.
- Not merely a prompt-engineering improvement.
- Not necessarily a proof-caching speed optimization.
- Not a claim of complete cloud-security correctness.

11. Provisional Research Questions and Hypotheses

11.1 Primary research question

How can persistent, dependency-aware invariant preservation be used to reduce unintended security and infrastructure regressions during iterative LLM-based Terraform remediation?

11.2 Secondary questions

- Does explicit invariant state reduce regression rates compared with stateless verification?
- Does resource-identity-aware validation reduce unintended Terraform restructuring?
- Can dependency-aware selective re-verification preserve correctness while reducing unnecessary verification work?
- What trade-offs does invariant preservation introduce in repair success, iteration count, latency and implementation complexity?

11.3 Candidate hypotheses

- H1: Stateful invariant preservation will reduce unintended regressions relative to stateless verification.
- H2: Resource-identity-aware acceptance will reduce unrelated resource mutations during iterative repair.
- H3: Dependency-aware selective re-verification can preserve protected properties while reducing unnecessary re-checks.
- H4: The added state/verification machinery will remain computationally feasible for a constrained Terraform subset.

12. Limitations and Threats to Validity

- Indexed-database coverage is incomplete because Scopus was inaccessible and not all planned database searches have a validated execution log.
- Several early screening decisions relied on abstracts or AI-generated summaries; those are not equivalent to complete full-paper review.
- Some named studies (especially IAMPERE and UPIR in the extraction stage) lacked sufficiently verified primary-source material in the provided evidence set and must not be treated as fully validated claims.
- Publication status varies between peer-reviewed papers, preprints and industry/vendor sources.
- The exact DAIL mechanism is still being designed; therefore novelty cannot be concluded from architecture sketches alone.
- The proposed invariant ledger may be unnecessary in some settings if full final-state testing already detects every relevant regression at low cost. This must be tested empirically rather than assumed.
- An invariant ledger that is poorly specified could itself become a source of false rejection or lock-in. Phase 3 must therefore include soundness and failure-mode analysis.

13. Implications for the CloudShield AI Product

The SLR does not replace the broader CloudShield AI product concept. CloudShield AI remains a larger cloud-security product that can include posture scanning, findings, context, AI explanations, remediation, reporting and automation. DAIL is the current research core inside that product.

CloudShield layer	Role
Detection / CSPM	Identify cloud/IaC security issues.
Context / risk	Explain affected resources, dependencies and severity.
AI remediation	Generate candidate Terraform changes.

CloudShield layer	Role
DAIL research engine	Preserve previously verified properties and control regression across iterations.
Verification / reporting	Present accepted/rejected fixes and evidence.

14. Phase 2 Final Decision

PHASE 2 STATUS: PASS — proceed to Phase 3, with DAIL treated as the leading research-gap candidate rather than a proven novel invention.

The review has achieved its main purpose: it eliminated the weak generic contribution, exposed the major direct and adjacent prior art, identified contradictory evidence, and narrowed the research question to a specific failure mode—loss of previously verified infrastructure properties during iterative LLM remediation.

14.1 What is locked for the next phase

- CloudShield AI remains the broader product.
- DAIL is the leading research core.
- The research problem is regression-aware, stateful Terraform remediation—not generic AI CSPM.
- Novelty and patentability remain open questions.
- Phase 3 must formalize the mechanism before any implementation claims are made.

14.2 What is deliberately NOT locked

- Exact data structure of the ledger.
- Exact invariant language/logic.
- Whether Z3/SMT is required.
- Exact selective re-verification algorithm.
- Exact benchmark/dataset.
- Patent claims or filing strategy.

15. Evidence Map and Key Sources Used in Phase 2

Source	Use in this report	Evidence reference
CloudShield Phase 2 thematic synthesis	User-provided synthesis; used for thematic structure, field evolution, verification strategies and evidence gaps.	[Source: Phase 2 thematic synthesis, pp. 1–3]
CloudShield stateful-invariant analysis	User-provided analysis; used for DAIL candidate mechanisms and closest-prior-art comparison.	[Source: Stateful-invariant extraction, pp. 9–10]
ConVer evidence	User-provided extraction; used to identify the strongest methodological adjacent threat.	[Source: Stateful-invariant full-text extraction, pp. 10–11]
TerraRepair evidence	User-provided full-text extraction; used for dependency context vs persistent state distinction.	[Source: Stateful-invariant extraction, pp. 1–2]
TerraProbe evidence	User-provided extraction; used for layered verification and deceptive-fix evidence.	[Source: Stateful-invariant extraction, pp. 2–3]
Evidence boundary report	User-provided verification report; used to constrain claims when full text was unavailable.	[Source: Evidence-boundary report, opening note]
Phase 2 gap synthesis	Derived from the reviewed Phase 2 evidence; the wording is intentionally provisional.	See Sections 9–14 of this report.

16. Handoff to Phase 3

Phase 3 should convert the provisional gap into an exact technical contribution. The next phase should define: (1) the Terraform subset and resource model, (2) the invariant representation, (3) the resource-identity model, (4) the dependency graph and invalidation rules, (5) the repair-acceptance algorithm, (6) baselines, (7) evaluation metrics, and (8) failure/soundness boundaries. Before implementation, the proposed mechanism should again be challenged against the latest literature and patent landscape.

Prepared as the consolidated Phase 2 research reference. Raw Gemini/Elicit reports remain supporting artifacts; this document is intended to be the primary working summary for subsequent phases.